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Institution: South East Technological University
Subject: SD3 Project
Purpose of Submission: Software Engineering 3 Project
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Blogger Manager – Iteration 1 Report

A page that helps people and newspapers to find nearby bloggers based on their location and topics they write about.

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Environment: Flutter + Firebase (Auth, Firestore, Hosting)

Deployment: <https://bloggermanager-f1e21.web.app>

Changes to the Vision

While the main focus is still about helping people and newspapers find nearby bloggers, the vision has been changed to specifically cater to the relationship between newspapers and bloggers as the two primary customers.

Changes to Specification

Additional Requirements:

- Implementation of the Google Maps API for geolocation.
- Creation of a Blogger Search dashboard for users to search for bloggers near them.
- Mandatory profile completion for new users before they can access the blog dashboard
- Requirement for web hosting deployment to test cross-device functionality.

Removed/Scrapped Requirements:

- Moderation Dashboard: Initialised but scrapped for Iteration 1 to focus on the main functionality of the application (Blog database).
- Direct Geopoint Storage: Attempted but replaced by Google Maps API integration due to storage issues in Firestore.

Existing Requirement Details:

- GDPR Compliance: Refined to ensure only general location (Town/County) is stored rather than pinpoint coordinates.
- Database Schema: Expanded to include more detailed blogger information, tags, and specific geolocation-linked metadata (title, description).

Changes to Research and Design

Research:

Shifted from passive learning (watching tutorials) to active development-based learning after finding that "mindlessly watching videos" did not sufficiently aid in troubleshooting Firebase.

Design:

Redesigned report and design specs to improve user experience (UX) for target customers.

- Redesigned the onboarding process to move away from single-page layouts toward a more structured login/sign-up flow.
- Refined UI cards and detail panes for better responsiveness on larger screens.

Goals for Iteration 1:

- ☒ ~~Look into unfamiliar tech and familiarise myself with Flutter, Dart, and Firebase (especially Firebase as it can also be used for the backend database)~~
- ☒ ~~Simple user interface to simply test the features of the blogger manager.~~
- ☒ ~~Defining the database schema to store blogger info, tags, and geolocation data~~
- ☒ ~~Setting up the API infrastructure to allow communication between Flutter frontend and the database.~~
- ☒ ~~Establishing the basic security and data privacy in the Blogger Manager required for handling user registrations. Ensuring we follow GDPR regulations.~~

Additional Goals achieved:

- ☒ ~~Successfully implemented Google Maps API to the application.~~
- ☒ ~~New Dashboard Implemented: Blogger Search~~
- ☒ ~~Ensured privacy safe location behaviour (no pin point location, just general overall location [e.g. Town, County].~~
- ☒ ~~Better onboard login/sign up experience, instead of lazily having everything on the same page or having everything on one page.~~
 - ☒ ~~new users are directed to complete a profile before dashboard access.~~
- ☒ ~~Deployed application to a web hosting service.~~

Use of Gen-AI Tools

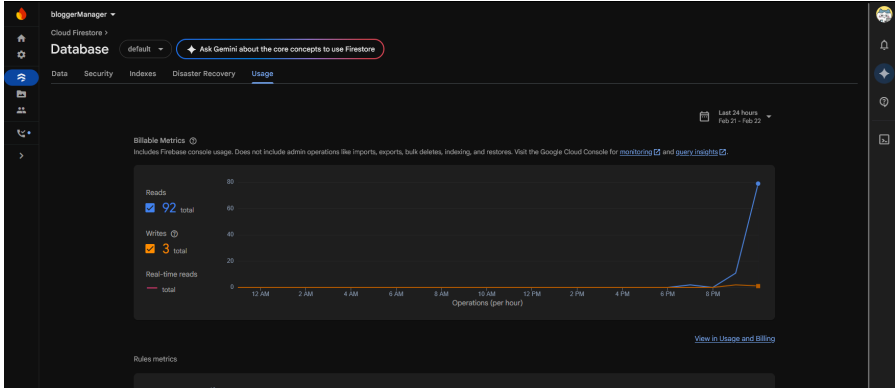
GitHub Copilot:

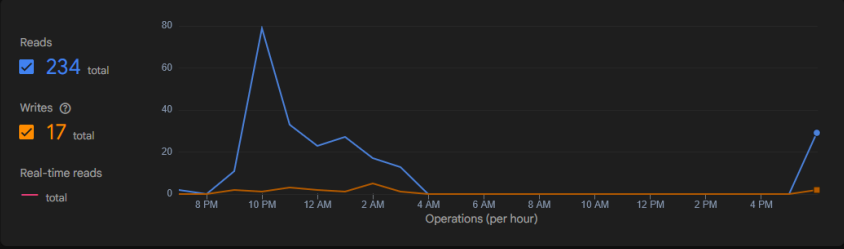
Utilized within VS Code to assist with coding tasks and speed up development. While AI is efficient for "vibe coding," it is prone to hallucinations; therefore, I familiarized myself with the tech stack to avoid being fully reliant on it. It was of help when I had stages of development where I had just hit a wall and couldn't progress any further.

It was of great use as I was able to further my understanding of Firebase to solve the issues that I struggled with for days during development. It also gave me step by step documentation on how to set up some features on Firebase.

Project Diary / Phases of the project

Date Range	Phase	Description
Feb 8 – 11	Studying Technology	Studying basics of Flutter Frameworks and Dart. Did this by reading Flutter documentation and watching tutorials on how to set up Flutter and its basics in Dart (<i>see Documents read, and Youtube creators I watched</i>). I needed this so that way I can speed up development, utilising the vscode github copilot, it can assist me with tasks, and I won't be confused when there is an error. But I cannot be fully reliant on AI, and as vibe coding may be beneficial and efficient, it is prone to hallucinations which may cause issues further into development. Which is why I needed to spend my time familiarising myself with my environment before I could begin development. Documents read: https://docs.flutter.dev/ Youtube creators I watched: Flutter Basic Training – https://www.youtube.com/watch?v=1xipgO2Wu8s Flutter & Firebase App Tutorials – https://www.youtube.com/watch?v=sfA3NWDBPZ4&list=PL4cUxeGkcC9j--TKldkb3ISfRbJeJYQwC Note: I need to improve my way of learning these technologies (especially Firebase), as further into development this has stunned my progress. Mindlessly watching videos did not benefit me whatsoever, and I need to do more development as you learn.
Feb 13 – 16	No Change. Break in research and development.	N/A
Feb 17 – 18	Setting up Database	Started by setting up a Firebase account, set up Firestore database. This was then later on removed and deleted in future phases as they will be initialised and created through code.
Feb 19 – 21	No Change. I had work.	(<i>Changed work schedule to prioritise project development and deployment</i>)
Feb 22	Trying to figure our Flutter -> Firestore	Conducted Flutter-to-Firestore database connectivity tests. This took a while to get working as I was still unfamiliar with Firebase. I made a simple User Input to Firestore Database application which took the users text that they've submitted and it was sent to the Firestore database.

	Firebase Hosting and Authentication services initialised.	Authentication for Sign Up and Login has been set up. A web host has also been initialised for deployment of my app to see if it works on other devices, and how my UX/UI looks on other devices.
	Dart Profile Dashboard Pages created	Created a Login/Sign Up page and a blogger dashboard. However, it was unsuccessful when attempting to connect it to the Firestore database.
	Flutter Geopoint Location and Moderation Dashboard initialised. Firestore database being stingy.	With the assistance from co-pilot, geopoint location was attempted. However, failing to store into the Firestore database, I decided to delete this and utilise the Google Maps APIs in future phrases instead as I needed to focus on the main part, Flutter -> Firestore. The moderation page was initialised, but for the sake of the first iteration, it was best for me to focus on the must have's of my application. I decided to scrap that and focus on configuring my Firestore database to allow information to be written within its firestores.
	Firestore database being stingy.	I have conducted many tests nonstop, and it only reads the database and not writes. The data being sent from the sign-in page is shown on the firebase authenticator, but not the firestore database. Development will continue the next day.  Figure 1. Firestore database usage graph showing a spike in reads due to tests, written data were done manually.
Feb 23	Retrying development.	Hit a wall in development due to Firestore Database. This has resulted with deleting pages, and re-initialising development building off of my simple Flutter -> Firestore web app I built yesterday.
Feb 23 (EXACTLY 18:26)	Successfully Fixed Firestore database issue!	It was due to an issue with security rules within the firestore database. You can see the small bump in writes.

		Billable Metrics ⓘ Includes Firebase console usage. Does not include admin operations like imports, exports, bulk deletes, indexing, and restores. Visit the Google Cloud Console for monitoring and query insights.  Reads 234 total Writes 17 total Real-time reads total Operations (per hour) Feb 22 - Feb 23 View in Usage and Billing Figure 2. Firestore database usage graph showing a small bump in writes. The flutter -> firestore finally works.
20:07	Successfully made a Login/Sign Up page.	Signing up now stores your email/user ID to the authenticator and to the firestore database. It took a lot of tests beforehand in order to reach this point see Figure 3, the solution was still in the firestore security tab see Figure 4. A simple Login user interface asks for the user's email and password which is then sent to the authenticator in Figure 4, and then sent to the firestore database (doesn't store password).

ResultsQuery Explain

Query results

Document Name	createdAt	displayName	email	lastLoginAt	profileText	testText	uid	updatedAt
"Users/3Q9HwjLeCFYFzpgVKMwqyM51J2"								
"Users/3AwXVAIPo4RnCbL7KNoUosm9ONM2"	February 23, 2026 at 8:01:52 PM UTC	null	"placeholder@email.com"	February 23, 2026 at 8:01:52 PM UTC	"12345"		"3AwXVAIPo4RnCbL7KNoUosm9ONM2"	February 23, 2026 at 8:01:54 PM UTC
"Users/TestingID"						"AAAAA"		
"Users/yfIADfOWObWQnIH35wK8ZwRCuI3"	February 23, 2026 at 8:04:58 PM UTC	null	"again@email.com"	February 23, 2026 at 8:04:58 PM UTC	"12345"		"eyfIADfOWObWQnIH35wK8ZwRCuI3"	February 23, 2026 at 8:04:58 PM UTC

Items per page: 501 - 4 of 4

Figure 5. Firestore database query

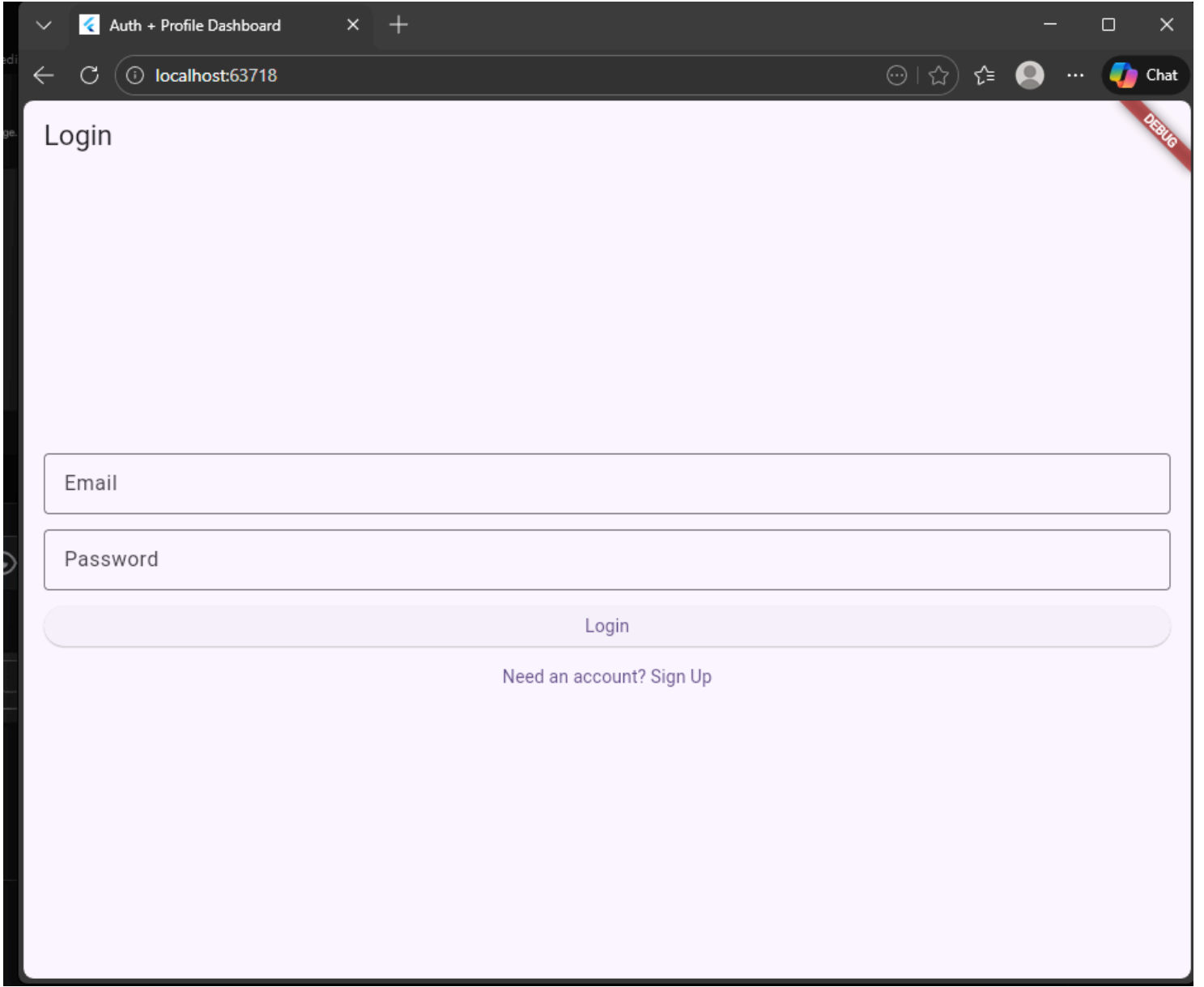


Figure 6. Log In Page (early stages)

	Simple profile dashboard created.	A simple dashboard has been created, showing up after a user has signed in. See Figure 7.
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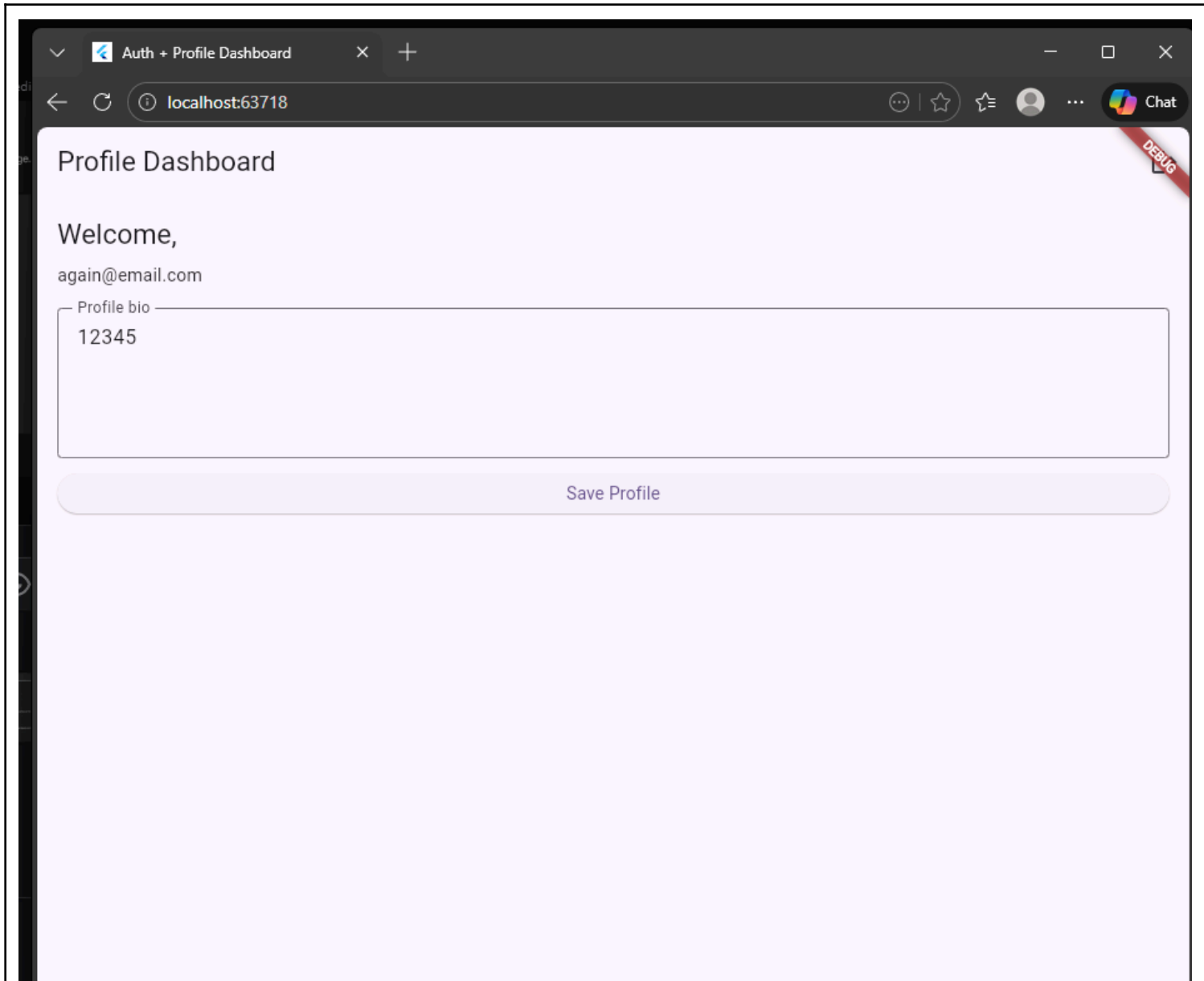


Figure 7. Profile Dashboard (early stages)

	Blogger Discovery (Initial)	Discovery features were added to browse/search bloggers with tag filtering and location aware logic.
Feb 24	Change in Research and Development Documentation	Redesigned design/report specs for improved UX. This is to mainly cater to the relationship between Newspapers and Bloggers. Those are our 2 main customers. Change in database schema to allow more blogger information and accurate geolocation using the Google Maps API. More on the Geolocation to be discussed in future phases.
DEVELOPMENT - FULL TIME 67 COMMITS TOTAL		
Feb 24	Blog Upload & Metadata	Blog upload/edit workflows were extended with title, description, tags, and geolocation-linked metadata. See Figure 8.
	UI/UX Card Refinement	Multiple cards and detail panes were redesigned for readability, spacing consistency, and responsive behavior on larger screens. This included improved profile and blogger presentation structure. See Figure 10.

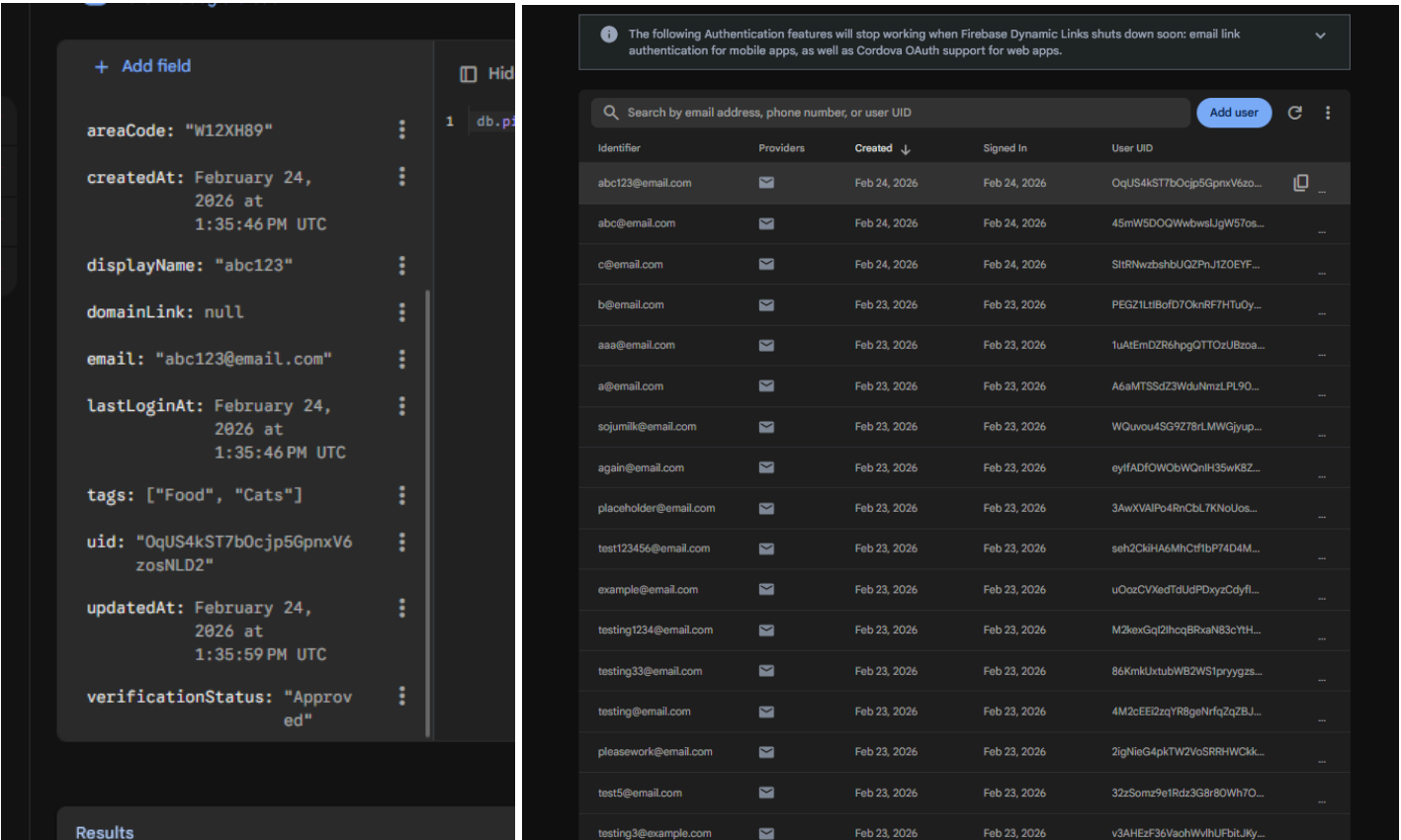


Figure 8. Blog Data in db. || Figure 9. Authentication Table of test users

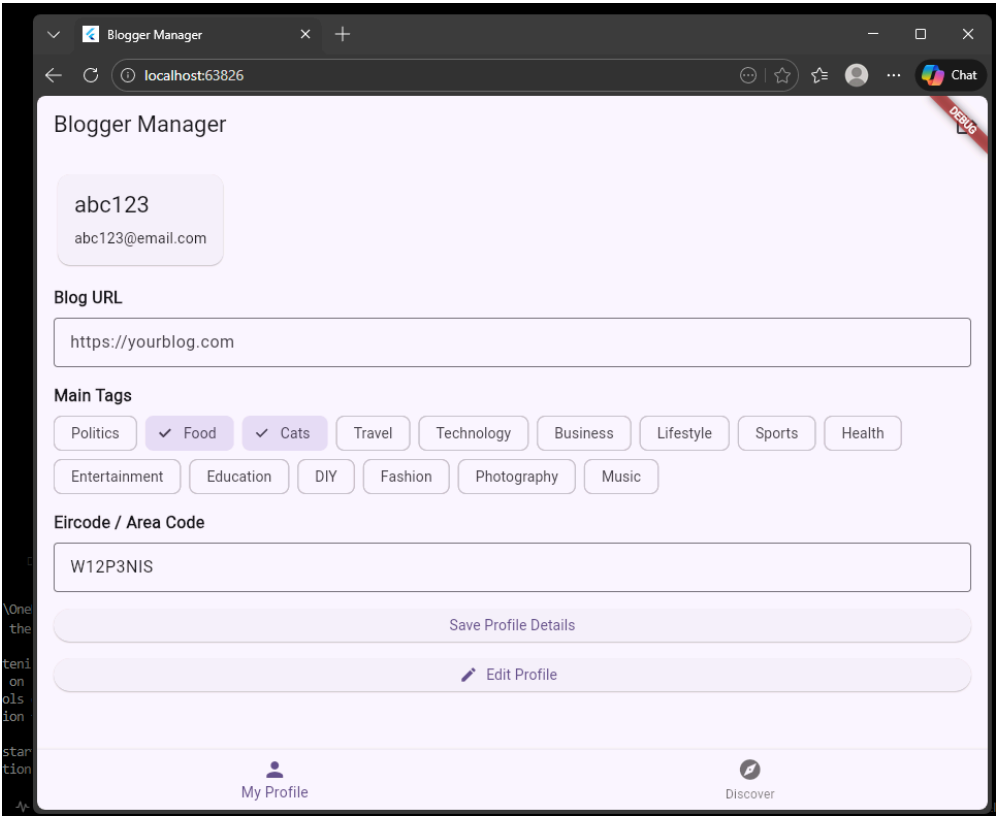


Figure 10. Profile Dashboard

Feb 25	UI/UX Pivot	Documentation overhaul; redesigned design/report specs for improved UX.
Feb 26	UI Refinement	Fixed RenderFlex overflows in Search/Detail screens.

		Auth UX Cleanup, Login/Sign-up screens were centered, padded, and have responsive card layouts with left-justified text input fields, and consistent spacing across device sizes. See Figure 11.
	Google Maps implemented.	Embedded Google maps onto the Home Page of the Blogger Manager. Also implemented within blogs so users can see where the blog has been uploaded in google maps. See Figure 11.
	Location & Geocoding	Location search and reverse geocoding were integrated to store city/county/country and support regional search behavior. UX feedback was added for permission and location failures, to help with debugging and to prevent any walls in development. See Figure 11.

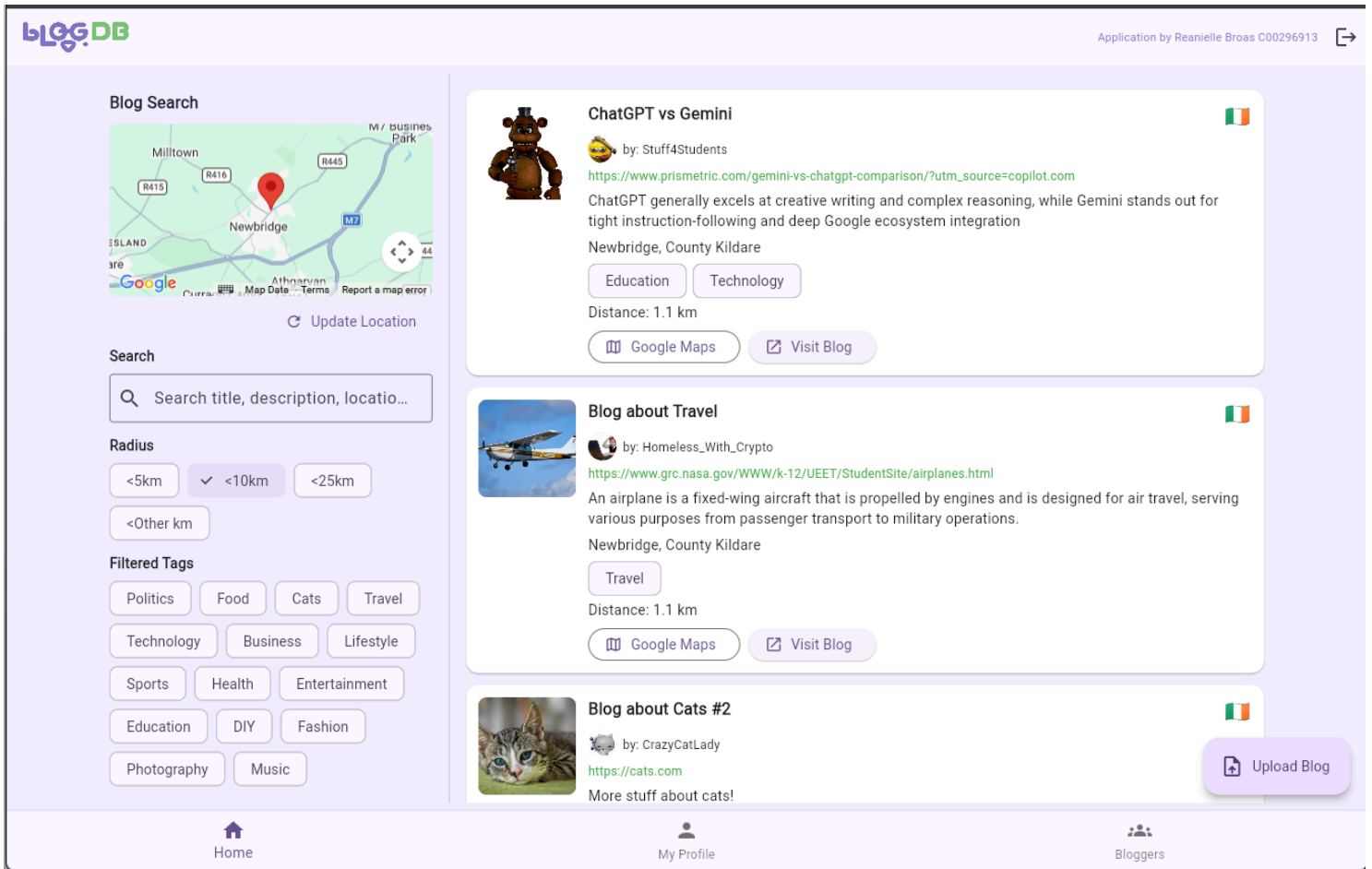


Figure 11. Home Dashboard

	Blog Row Visual Upgrade	“Blogs by this Blogger” items were rebuilt with image-left/details-right composition, with a bottom right button alignment, improving scannability and visual hierarchy. See Figure 12.
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	Country Icons on blogs!	Blog cards were updated to replace bottom-right country text with top-right flag icon indicators (with fallback behavior). See Figure 14.
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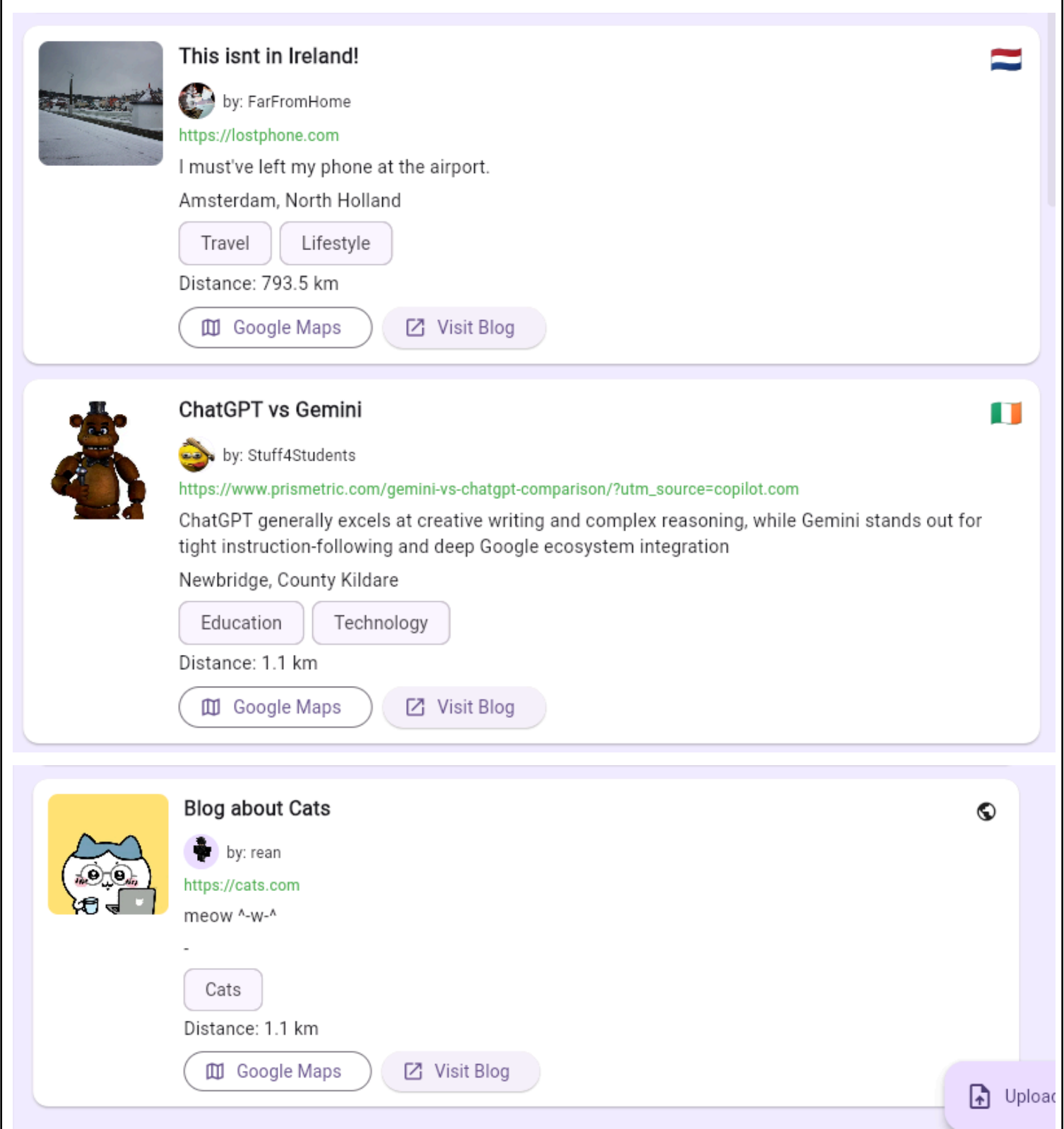


Figure 14. Row of Blog Cards with country icons at the top.

	Auth Routing Finalization	Auth routing logic was hardened so existing users with complete profiles bypass setup on login, while new signups are directed into completion flow until required profile data is finished.
	Static Validation	Repeated flutter analyze runs were completed after each major change set. Results remained clean, supporting release confidence and reducing regression risk before deployment.

Feb 27	Fixes & Deploy	Resolved <code>MissingPluginException</code> ; finalized Firebase Hosting deployment. Switched from <code>image_input</code> to <code>file_input</code> . I had friends to test out the application to see if it worked on their device.
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Goals Vs Outcomes

Goal	Outcome	Validation
Tech Familiarization	Familiarised with Dart, Flutter, and Firebase Auth/Firestore.	Build was successful and deployed through firebase hosting.
Database Schema	Defined schema for bloggers, tags, and geolocation.	Information is stored in the firestore database.
GDPR Compliance 1	Switched to general area queries (City/Country). Password	Tested at Home, with Friends, and with VPN. Now only shows the City/Country of a blogger instead of their pin-point location.
GDPR Compliance 2	With Firebase, a salted one-way script hashing algorithm ensures user passwords are never stored in plain text and remain unreadable.	Not shown in Firestore database. Can not be shown on Firebase authenticator, admins can only remove but not view passwords.
API Infrastructure	Established communication between Flutter and Firebase (after non-stop trial and error)	Live data fetching as seen in the Firestore database usage graph.
UI Testing	Created Blogger Manager dashboards (Discovery/Upload/Search).	flutter analyze passed

Revised Iterations Plan

My work schedule was modified to prioritize project development and deployment over other commitments. However, despite this commitment, I was privileged enough to have a work-life-university balance. Committing hours to this project, and setting aside days for work , and relationships.

Future phases will move toward more advanced geolocation features using the Google Maps API. I am thinking of having bloggers appear on the Google Map, in a way that doesn't leak their pinpoint location.

To enforce more safety, I will need to implement the moderation dashboard that I had scrapped. To prevent any ill use.

As I am using the firestore database, I will need to optimise my code to prevent any multiplications in reading as I am using a free plan for my database and hosting. I do not want to pay for more database storage.

Usage of "flutter analyze" runs after major changes to reduce regression risk before future deployments.